



Global CO₂ Emissions

Trends, Regional Shifts,
and Per Capita Inequalities

1750–2024 | Data from Global Carbon Budget 2025 & NASA

For policy analysts and climate negotiators



Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million

NOAA Mauna Loa Observatory
February 2026

Pre-industrial baseline

~280 ppm

Today

429 ppm

+50% increase

The current level is at least 1.5× greater than the natural increase observed at the end of the last ice age 20,000 years ago.

A change that took thousands of years then has occurred in roughly two centuries, driven primarily by the burning of coal, oil, and natural gas.

Source: NOAA / Mauna Loa Observatory, Global Carbon Budget 2025

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Bt to 35+ Bt



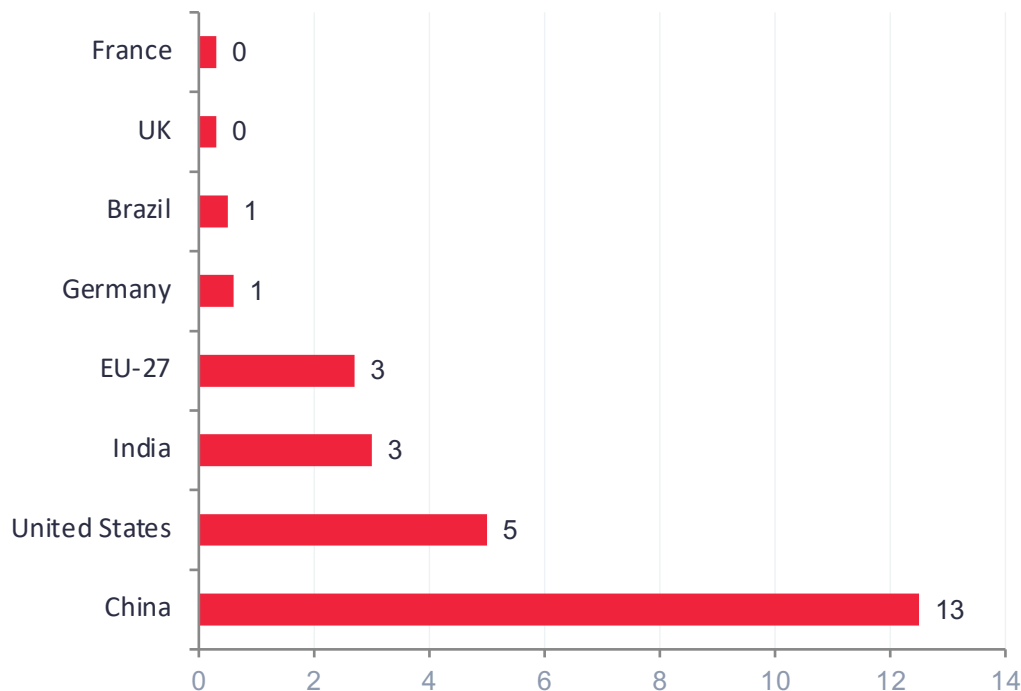
Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained relatively slow until the mid-20th century.

While emissions growth has slowed in recent years, global annual emissions from fossil fuels and industry have not yet peaked.

► See [figure_2.jpg](#) for the full emissions trajectory 1750–2024

Source: Global Carbon Budget 2025, Our World in Data

China Now Emits Over 12 Billion Tonnes — More Than Double the US



► See figure_3.jpg for annual CO₂ emissions by country

Key Insights

China

~27% of global total, slight increase

United States

~14%, declining from ~6 Bt peak (2005)

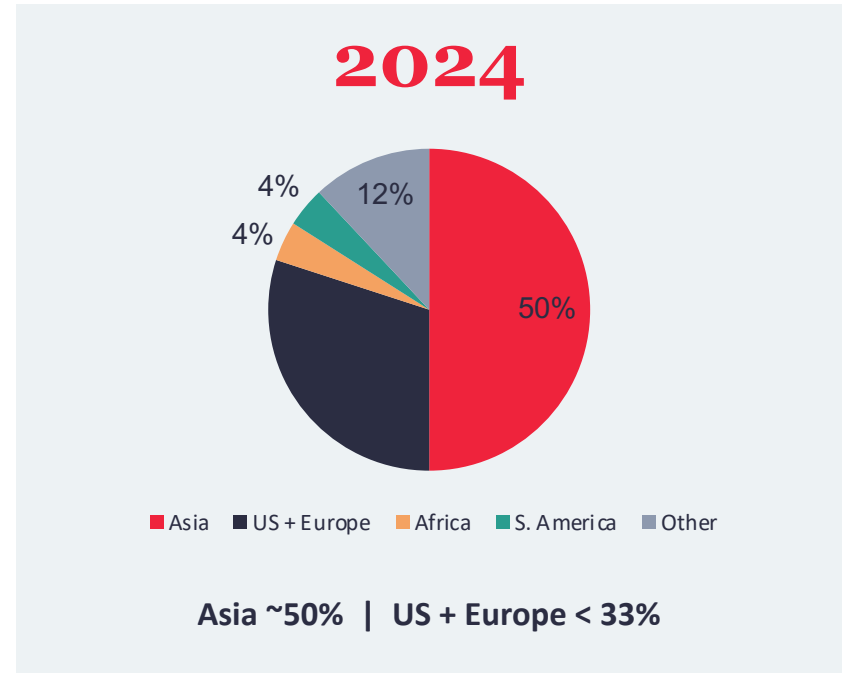
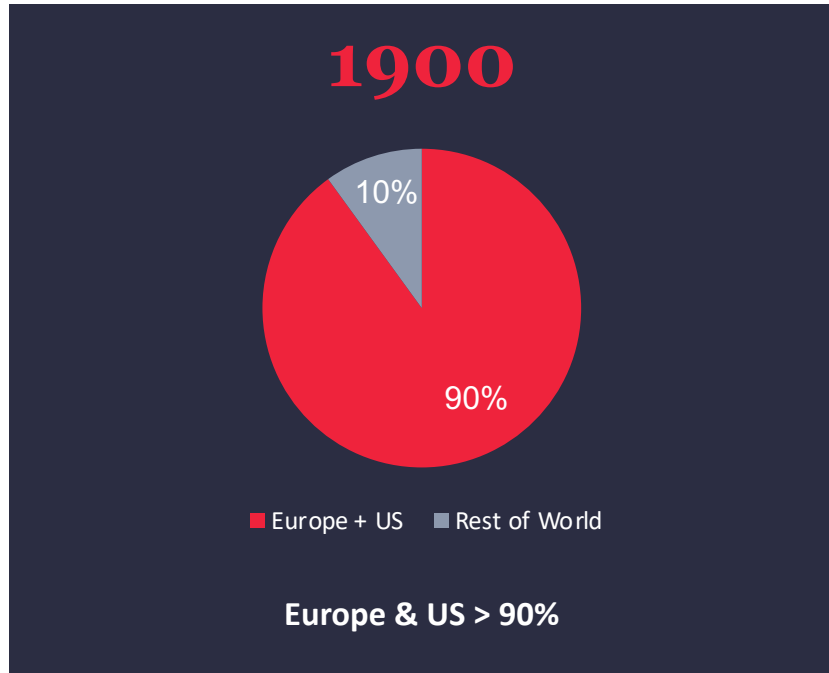
India

~8%, rising rapidly

EU-27

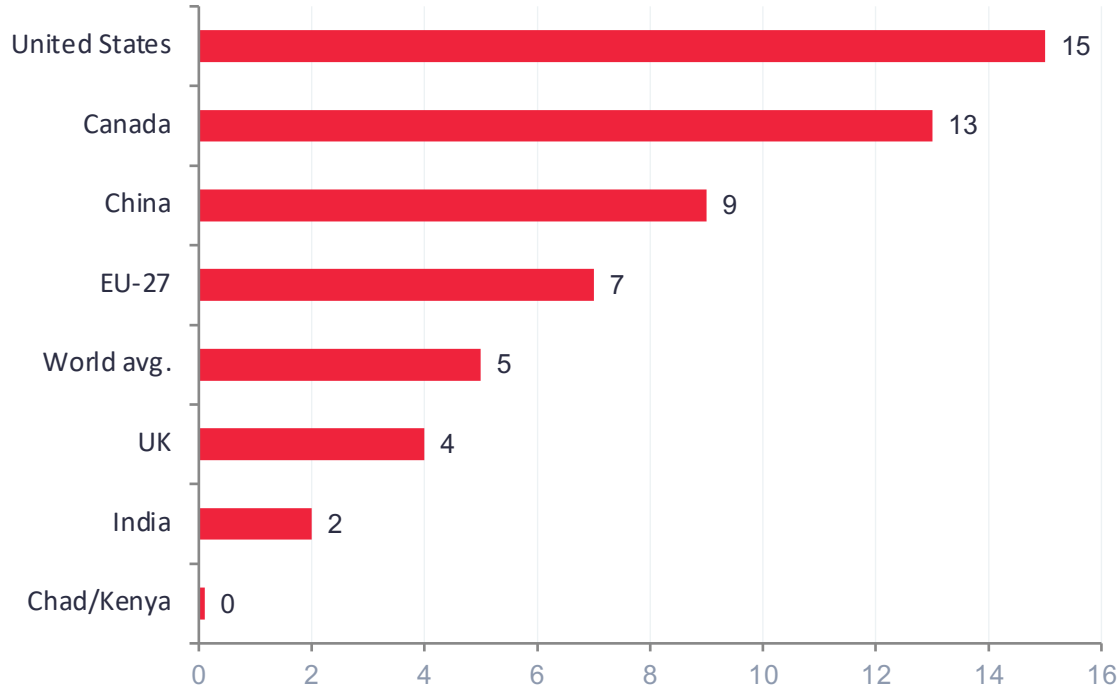
~7%, declining steadily

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



Africa and South America each contribute only 3–4% — comparable to international aviation and shipping combined.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



► See figure_1.jpg for per capita CO₂ emissions over time

150×

The average American emits in under two days what someone in Chad emits in an entire year.

Highest per capita emitters globally:

Qatar, UAE, Bahrain, Kuwait

Among large nations: US, Australia, Canada at ~3× world average

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

DECLINING (–1% to –5%)

United States –1% to –3%

Much of Europe –1% to –5%

United Kingdom Continued decline

INCREASING (+2% to +10%+)

Russia +2% to +5%

Parts of SE Asia +2% to +10%

Several African nations +2% to +10%+

Some S. American nations >+10%

The 2024 map shows a mixed global picture — the geographic pattern underscores diverging emission trajectories between developed and developing economies.

► See [figure_4.jpg](#) for annual percentage change in CO₂ emissions, 2024

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their energy sources.

	France	UK	Germany	Netherlands
Per Capita CO ₂	~4 t	~4 t	~8 t*	~8 t*
Nuclear Share	High	Moderate	Low	Low
Renewables	Growing	Strong (wind)	Growing	Growing
Fossil Fuel Share	Lower	Lower	Higher	Higher (gas)
Result	Lower CO ₂	Lower CO ₂	Higher CO ₂	Higher CO ₂

* Germany's per capita figure (~8 t) is estimated from its ~0.6 Bt annual total for ~84M population; Netherlands is similar in range.

Policy and energy mix choices — not just prosperity — determine per capita outcomes. Nuclear and renewables significantly reduce emissions intensity.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01 Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

02 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.

03 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.

04 Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).

05 Energy mix and policy choices matter: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions.